

I. Introduction

What makes car-classification important in the real world is the value of drone/camera surveillance. In the real world, drones and cameras are used to capture footage of important areas to help cut down on crime. One of those features of a surveillance system is to identify cars and license plates to track down potential criminals and monitor their whereabouts within a surveilled area.

Our goal is to train a model to accurately identify a car's make based on 100 test images, 20 validation images and 2 test images per car make. Most of the validation images and all test images were taken with a drone while the other images were stock images. Our car makes assigned to train our model on were Jeep, Kia and Volkswagen.

II. Data Collection

The data we collected was 122 images of Jeeps, Kias, and Volkswagens. This breaks down to 100 test images, 20 validation images, and 2 test images for Jeeps, Kias, and Volkswagens. The 100 test images for each car make were web-scraped stock photos. Most of the 20 images for each car make were photos taken from a drone, and the others were web-scraped stock photos. Both test images from each car make were taken from a drone. A table shows the split of each car make and image types.

Car Make	Jeep	Kia	Volkswagen
# of Train Images	100	100	100
# of Validation Images	20	20	20
# of Validation Images	2	2	2

Two samples of each car make are included as follows.

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III. Dataset Organization

Our folder structure is as follows.

Drones Group Project 1 Photos/

```
└─ Jeep_Kia_Volkswagon_Dataset/  
  └─ train/  
    └─ Jeep/  
    └─ Kia/  
    └─ Volkswagen/  
  └─ valid/  
    └─ Jeep/  
    └─ Kia/
```

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```
|   └─ Volkswagen/  
└─ test/  
    └─ Jeep/  
    └─ Kia/  
    └─ Volkswagen/
```

The train folder has 3 subfolders, Jeep, Kia and Volkswagen. Each subfolder has 100 images each. The valid folder has 3 subfolders, Jeep, Kia, Volkswagen. Each subfolder has 20 images. The test folder has 3 subfolders. Each subfolder has 2 images. That makes a total of 300 images in the test folder, 60 images in the valid folder, and 6 images in the test folder, which gives you a grand total of 366 photos in our dataset. Duplicates were avoided by going row by row in search results and scrolling down each time a row was exhausted in order to prevent downloading duplicate images.

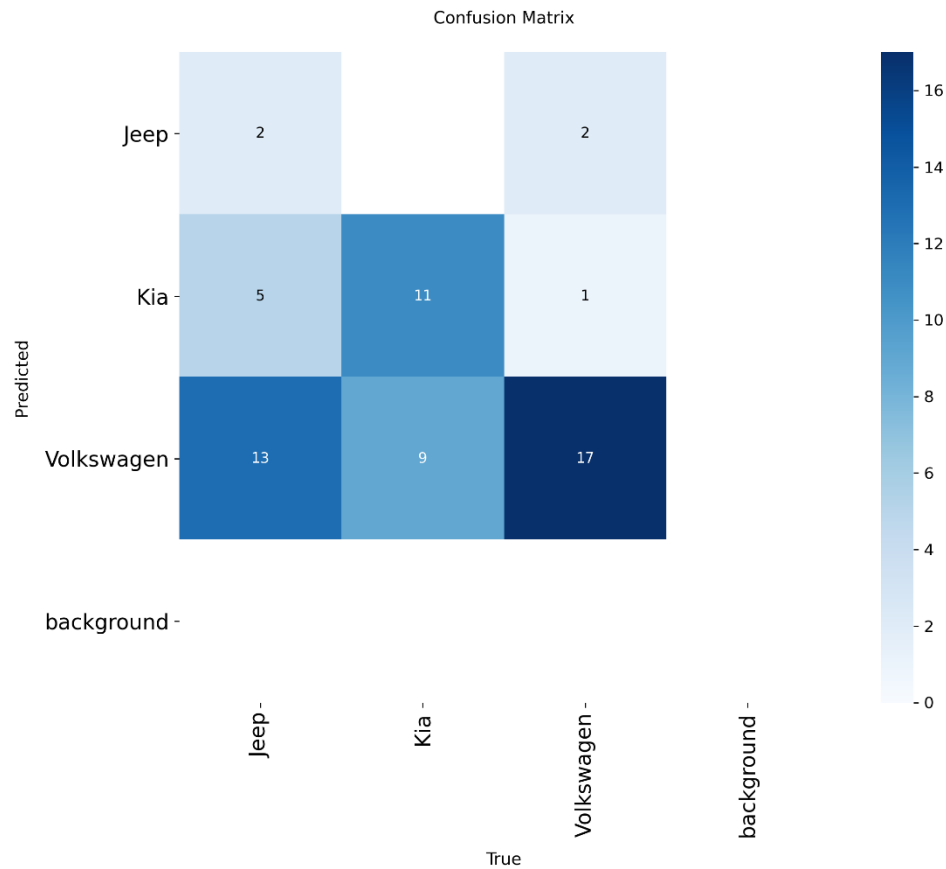
IV. Model Training

We used the YOLOv11 model variant 'yolo11n-cls.pt'. As for the training setup, we used 30 epochs, default batch size, and various image sizes. Most were between 500x500 and 1000x1000 image sizes. The training process went smoothly. The challenge we've overcome was a result with very low accuracy. We learned that training the model on just stock images wasn't enough. Next, we swapped out five of our drone images from the validation sets and put them in the train set while taking the five stock images from the train set and putting them into the validation set to make up for the five images moved to the train set.

V. Results & Interpretation

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Model #1 Results:



Best Top-1 Accuracy: 0.5

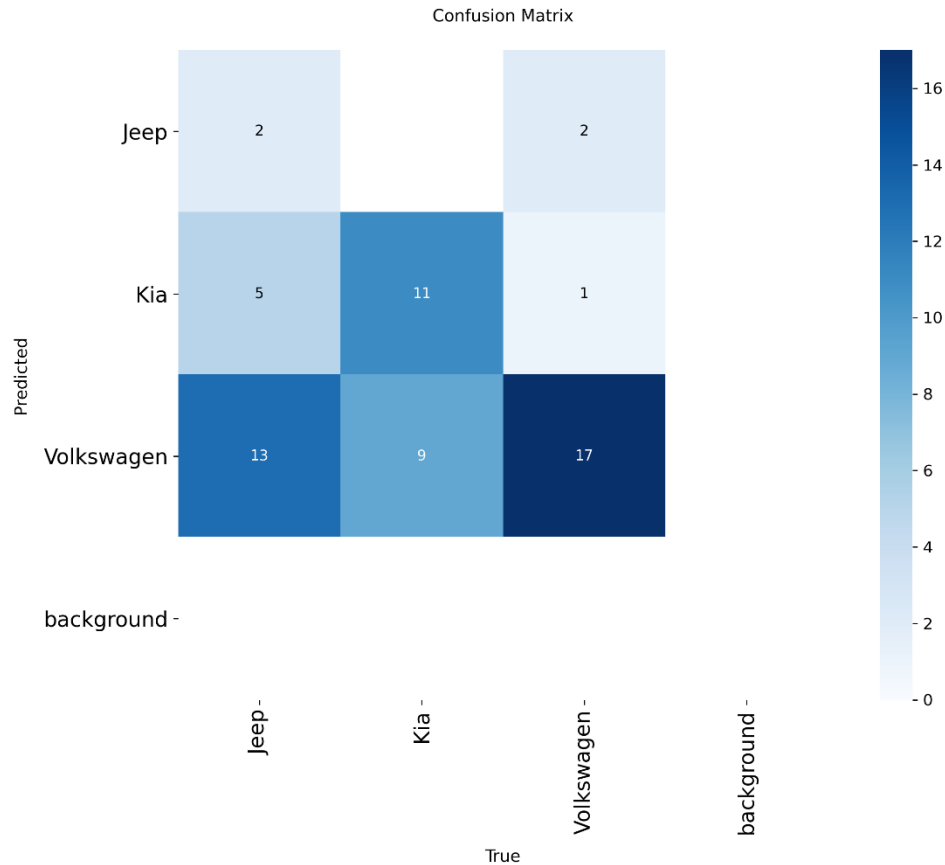
Minimum Training Loss: 0.02348

Minimum Validation Loss: 1.52688

After switching 5 images from each car make from their train folder with their validation folder counterparts, the second model performed much better.

Model #2 Results:

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Best Top-1 Accuracy: 0.75

Minimum Training Loss: 0.03295

Minimum Validation Loss: 1.02744

The confusion matrix of the second model shows that Jeeps were able to be almost 100% accurate while Kia's accuracy stayed the same. Something else to take note of is that the Best Top-1 accuracy improved significantly by 0.25, and the minimum validation loss improved significantly by about 0.5.

Overall, the results showed that instead of training a model on just stock photos, we should include some of the drone images to train our model on so it can identify car makes more accurate.

VI. Reflection & Next Steps

Looking back at the results and the photos, there were plenty of things we could've done to improve our model's accuracy and overall performance. We should've taken more images with the drone and relied less on stock images when it came to training the model. That would have been the singlemost improvement to accuracy. Another thing we could've done was optimize the angle at which the photos were taken to give the model an easier time detecting the car make. And lastly, taking photos at a different time of day could've given us better lighting. A sunny day is nice, but not with all the tree shadows interfering with image quality.